Soubhik Sadhu

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Summary

Final-year Computer Science Engineering student with strong foundations in Data Structures and Algorithms, Object-Oriented Programming, DBMS, and Computer Networks. Backend-focused developer experienced in building scalable systems and RESTful APIs using FastAPI and MySQL with performance optimization.

Education

SRM Institute of Science and Technology — Chennai, Tamil Nadu
Bachelor of Technology in Computer Science and Engineering (AI & ML) — *August 2023 – May 2027*

- CGPA: 9.19/10.0
- Relevant Coursework: Data Structures and Algorithms, Machine Learning, Computer Vision, Database Management Systems (DBMS)

Experience

AI/ML Intern — December 2025 – January 2026
Eastern Coalfields Limited (ECL), A Subsidiary of Coal India — *Asansol, West Bengal*

- Built **ECL-SalarySight**, a full-stack payroll ingestion system (FastAPI, React, MySQL) to replace a manual Excel workflow, handling dynamically structured monthly salary sheets with varying layouts
- Designed a dynamic Excel-to-MySQL ingestion pipeline with upsert logic keyed on a composite primary key (employee number + date/month + area), guaranteeing 100% duplicate-free records across 5,000+ entries/month
- Optimized REST APIs to process 10,000+ records in under 5 seconds with API response latency under 200 milliseconds

Projects

SkillSync | *FastAPI, React, PostgreSQL, BERT, FAISS* | **GitHub** — February 2025

- Engineered the backend architecture for a 6-module Career Intelligence Platform using FastAPI and PostgreSQL, serving features like a Core Recommender and user DSA Tracker
- Applied BERT (bert-base-uncased) embeddings with cosine similarity for semantic skill matching, combined with a blended scoring formula (60% cosine similarity, 25% profile score, 15% DSA score) to rank job matches
- Implemented FAISS (Facebook AI Similarity Search) indexing to optimize embedding retrieval speed, enabling sub-second similarity search across the job recommendation pipeline

OcuSight | *Python, PyTorch, FastAPI, React, TypeScript* | **GitHub** | **IJIRT Published** — March 2026

- Evaluated ResNet-50, EfficientNet-B4, and DenseNet-121 on RFMiD dataset (3,200+ fundus images, 46 retinal conditions); achieved 97.8% per-label accuracy with BF16 mixed-precision and 15ms inference on RTX 4060
- Designed a confidence-aware ($\geq$70%) LLM pipeline via Gemini API for clinical insights and specialist referral guidance; deployed end-to-end diagnostic platform using FastAPI and React TypeScript
- Published in IJIRT: *"OcuSight: Benchmark Performance vs. Clinical Deployment Reliability in Retinal Multi-Disease Classification"*

AgroWare | *Python, PyTorch, FastAPI, React, PostgreSQL, TF-IDF* | **GitHub** — January 2026

- Built a plant disease detection pipeline (25 classes, 5 crops, 15k-image dataset) benchmarking four CNN backbones (ResNet50, MobileNetV2, EfficientNet-B0, DenseNet121); deployed EfficientNet-B0 at 82.4% accuracy
- Created a crop recommendation system (87.4% accuracy) with ensemble ML models, automated weather/soil data fetching, and explainable AI with parameter-level justifications
- Developed a multilingual farmer-centric chatbot with TF-IDF-based intent matching, supporting English, Hindi, Punjabi and Hinglish using curated government agricultural datasets

Technical Skills

Programming Languages: Python, C, C++, JavaScript, TypeScript, SQL
Machine Learning & AI: PyTorch, TensorFlow, Scikit-Learn, NLP, BERT, FAISS, Deep Learning
Web Development: FastAPI, React, RESTful APIs, HTML/CSS
Cloud & DevOps: AWS (EC2, S3, CloudWatch, IAM), Docker, Kubernetes, Railway
Tools & Databases: Git, GitHub, PostgreSQL, MySQL, OpenCV, Jupyter Notebook

Achievements

Completed Amazon ML Summer School (MLSS) 2026

Published research paper in IJIRT (Volume 12, Issue 12, May 2026)

Creative Head (Events & Logistics), SIGKDD ACM Student Chapter - spearheaded logistics and execution of technical symposiums and hackathons (including HisenHack 2025), scaling events to engage 2,400+ participants